

Topology First-Year Exam, January 2020, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that the set of functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ that are eventually 0 is countable.
2. Let $A \subset \mathbb{R}^2$ be a countable subset. Show that $\mathbb{R}^2 \setminus A$ is connected.
3. Show that every continuous map $f : [0, 1] \rightarrow [0, 1]$ has a fixed point. Is this true for continuous maps
 - (a) $f : [0, 1] \rightarrow (0, 1)$?
 - (b) $f : (0, 1) \rightarrow (0, 1)$?
4. Let X be connected, show that
 - (a) X^n is connected;
 - (b) X^ω is connected.
5. Let $f : X \rightarrow X$ be a map of a compact metric space to itself that satisfies the following condition:
 $d(f(x), f(y)) < d(x, y)$ for $x \neq y$.
 - (a) Prove that f is continuous.
 - (b) Show that f has a fixed point and the fixed point is unique.

Answer the following with complete definitions or statements or short proofs.

6. State the Intermediate Value Theorem.
7. Does there exist a continuous surjective map from the 2-sphere S^2 to the interval $(0, 1)$?
8. Does there exist a continuous injective map from the 2-sphere S^2 to the interval $(0, 1)$?
9. State the Cantor–Schröder–Bernstein Theorem.
10. Is every connected space path connected?
11. What is a basis of a topology? Does the set of all half-open intervals $\{(a, b], [c, d) \mid a < b, c < d\}$ form a basis of a topology on $\mathbb{R}$?
12. State the Extreme Value Theorem.
13. Is $\mathbb{R}^\omega$ connected in the uniform topology?
14. Is the set of integers $\mathbb{Z}$ locally compact? Is the set of rational numbers $\mathbb{Q}$ locally compact?
15. Is $\mathbb{R}^2$ with the dictionary order topology metrizable?